

25th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems

Heterogeneous ensembles of classifiers in predicting Bundesliga football results

Jan Kozak^{a,*}, Szymon Głowania^a^a*Department of Machine Learning, University of Economics in Katowice, 1 Maja, 40-287 Katowice, Poland.*

Abstract

Predicting sports results is a well-known problem, and it applies to the more predictable individual sports as well as to the less predictable team sports – e.g., football. In the vast amount of information and parameters that are associated with team sports, there are few solutions related to the comprehensive approach of knowledge discovery in databases (KDD). In this paper, we propose a new approach to the whole process of discovering knowledge from data to predict the result of Bundesliga football matches. The main goal of the paper is to improve the prediction of football match results using heterogeneous ensembles of classifiers. To achieve the goal, it is necessary to prepare a training set and to develop a learning model – in this case, we analyze whether the use of limited information allows for the correct prediction of the results of football matches. The use of the developed model based on heterogeneous ensembles in the prediction of Bundesliga match results allows for improved prediction compared to classical ensembles models and single classifiers. Additionally, the possibilities arising from the use of heterogeneous ensembles of classifiers allow to distinguish some of the analyzed cases. The proposed approach was tested on real-world data, the results of Bundesliga football matches analyzed from the 2010/2011 season to the 2019/2020 season. Experiments were performed on methods selected based on an in-depth literature review. Conducted experiments, supported by the proper statistical analysis, confirm the usefulness of the proposed approach in predicting Bundesliga football results.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of KES International.

Keywords: heterogeneous classifiers; ensembles of classifiers; sport result prediction; Knowledge discovery

1. Introduction

The increase in computing power of computers had a positive impact on the possibilities of using artificial intelligence in various disciplines of life. Machine learning as one of the available approaches is an increasingly used tool in the process of knowledge discovery in databases (KDD). The great popularity of sports competitions and the huge financial resources associated with them cause an increased interest in the possibility of using machine learning

^{*} Corresponding author.*E-mail address:* jan.kozak@ue.katowice.pl

algorithms to predict the results of these games. However, the literature still lacks solutions dealing with the subject of prediction of results with an appropriately comprehensive approach based on KDD.

In the available papers, one can find analyzes on the use of machine learning to predict results in various sports, including football [1, 10, 14, 25, 26], basketball [21], rugby [19], javelin throwing [18] or American football [6]. At the moment, the largest collection of studies concerns football predictions for the English Premier League (this is due to the availability of ready-made data sets), while other less popular leagues or leagues of other countries leave a wide field for research. When selecting an appropriate machine learning algorithm, a common approach is to compare with each other (in different combinations) the results obtained by decision trees, support vector machine, and random forests, so the proposed approach will be compared with the results achieved by these algorithms.

Classical ensembles of classifiers consist of weak classifiers, and the prediction process is performed by voting each classifier. The first approach proposed in the literature is based on creating homogeneous ensembles of classifiers, an example of which is bagging, AdaBoost or random forests [15]. The next step to improve the results is to create heterogeneous ensembles of classifiers based on many different models, so they can better represent the process and consequently provide a more accurate classification result.

On the basis of the presented problem analysis, the main goal of this paper is to improve the prediction of football match scores using heterogeneous ensembles of classifiers. To verify the proposed approach and test whether it allows improving the classification results, it is necessary to develop a training set and learn individual models.

Moreover, we hypothesize that the application of the developed model based on heterogeneous ensembles in the prediction of Bundesliga match results allows for improved prediction compared to classical ensembles of classifiers and single classifiers. Additionally, the approach based on the prediction of only selected football matches (in which the decision of all classifiers is uniform) allows for improvement of prediction.

The article is organized as follows: In the next section, we present the theoretical background of the research related to the algorithms used in this paper and the problem studied. The literature review is discussed as well. The third section is focused on predicting football results based on proposed heterogeneous ensembles of classifiers. The fourth section includes the numerical experiments and the statistical verification of the generated results. The last sections include a short summary and details of future research.

2. Background

The problem of classification of sports results is critical important, which was confirmed by numerous publications on this subject. In addition, machine learning algorithms were often used in this field, so we analyzed both the problem of prediction of results as well as the theory in the field of machine learning algorithms.

2.1. State of the art

The high popularity of sporting competitions and the associated business is causing an increase in the popularity of methods for analyzing and predicting the results of sports meetings. One of the most frequently analyzed sports is football. Among the most popular league is the English Premier League, for which data sets with many of attributes are prepared and available. However, among the sports analyzed using machine learning, there are also basketball, rugby, javelin throw or American football.

2.1.1. Predicting football results

The paper [1] broadly covers the subject of prediction of football match results, the number of goals scored and the process of selecting appropriate variables for the model. The authors point out that the English Premier League is characterized by high variability, which also has a large impact on the prediction of results.

R. Baboota and H. Kaur cite other key research results, including:

- use of the Bayesian network by A. Joseph, N. Fenton and M. Neil [14];
- use of Monte Carlo sampling with Markov chains in conjunction with a Bayesian network by H. Rue and O. Salvesen [24];
- use of fuzzy logic by A. P. Rotshtein, M. Posner and A. B. Rakityanskaya [23];

- use of social media data and traditional statistical techniques by R. P. Schumaker and T. A. Jarmoszko and C. S. Labedz [27].

The studies conducted by the authors concerned the possibility of using, among others: Gaussian naive Bayes, support vector machine, random forest, and gradient boosting. The highest accuracy achieved in the set of these methods was obtained for gradient boosting and random forest algorithms.

In [14] the authors have made predictions of football match results for Tottenham Hotspur Football Club. They paid considerable attention to the use of, among others, decision trees, Naive Bayesian and a K-nearest neighbors. The study was based on match results from 1995 to 1997.

G. Schauburger and A. Groll published in [25] a study on predicting the results of football matches. Their experiment was based on the results of the FIFA World Cup from 2002–2014. They used random forests in prediction, and the classification was based on two approaches. The first one made predictions between win / draw / loss. The second one predicted the exact number of goals scored.

The findings of E. Eryarsoy and D. Delen were presented in [10]. The authors made an analysis based on the Turkish Super League (results from 2007 to 2017) concerning the prediction of football match results. The prediction was made on two levels: win/draw/loss and: team will score / team will not score. Special attention was paid to algorithms: random forest, decision trees, naive Bayes, ensemble methods, support vector machine, k-nearest neighbors, and neural network. The highest accuracy on the examined set was achieved by random forests.

G. Schauburger, A. Groll and G. Tutz in their work [26] analyzed the possibility of predicting the result of a match using three variants of the Bradley-Terry model, which they developed accordingly to improve the result. In addition to applying a statistical approach to prediction, the authors' goal was to identify the main variables influencing team success. Authors analyzed the collected and possible to use variables regarding individual team statistics related to, for example, the percentage of ball possession or the effect of games at their own stadium. This is one of the few papers including the research on the data on football matches (2015/16 season) of the German Bundesliga. The accuracy obtained in the article exceeded that of the bookmaker and was on the level of 42–50%.

2.1.2. Predicting other sports meeting results

Despite the significant predominance of publications on the analysis of the possibility of using machine learning in the prediction of football meets performance, studies on other sports can also be found.

The study on predicting the results of basketball matches was prepared based on the use of the support vector machine algorithm. The results of the trained model were compared to the decision tree models and showed greater accuracy [21].

The most common approach of selecting the appropriate machine learning algorithm is the comparison of results obtained by decision trees, support vector machine, and random forests with each other. In most studies, the latter of the above-mentioned approaches achieves the greatest accuracy, however, as can be seen in the analysis of the possibility of predicting the result (win / lose) for American football matches, it was presented in [6] that for the collected data the decision trees achieved the greatest accuracy (83%).

As presented in [17] during college football prediction, the authors of the new approach achieved a significant advantage over standard solutions related to neural networks, support vector machine and decision trees. This approach was based on a 4-dimensional space with the characteristics of each team and comparing the current pair to the most similar case from historical data.

2.2. Machine learning algorithms

Classification as one of the methods that allow data mining involves creating a model to assign a new object to one of the predefined classes [15]. The following algorithms can be distinguished among the most popular classification methods, the following can be distinguished: decision trees, support vector machine, ensemble methods, neural networks, and Bayesian networks.

2.2.1. Decision trees

Decision trees are algorithms for classification with a tree-like structure with branches and nodes as main elements. Decision trees are directed acyclic graphs, nodes are made of vertices, and branches are connections between them.

Each leaf is a label for its category, while a node is a test of conditional attributes, and an edge is the result of the test. The root of a tree is a vertex without a parent, while vertices without a child are leaves. Many algorithms can be found in the literature to create decision trees including ID3, C4.5, C5.0, or CART [15].

The Classification and Regression Trees (CART) algorithm (chosen to be tested in this work) was developed by L. Breiman et al. in 1984 [4]. Two splitting criteria have been proposed for this algorithm: Gini and Twoing criterion. Using the Gini criterion, a test is searched for which will best divide the data in the node into two, maximally homogeneous parts. It is based on the Gini index, which is a measure of the concentration of a random variable. Using the Gini criterion, a test that will derive the best division of data in the node into two, maximally homogeneous parts is searched. It is based on the Gini index, which is a measure of the concentration of a random variable. The homogeneity of sets comes to the background [4].

2.2.2. Support vector machine

Support vector machine (SVM) is a machine learning technique whose goal is to classify a new element into one of the predefined classes. A training set with labeled instances is used for training. The created model contains information about the boundary separating the cases to each class. The main task of the algorithm is to find a hyperplane that optimally separates the cases into different classes. Striving for the optimum is achieved by obtaining the widest possible margin between classes which is associated with increasing the distance of points from the hyperplane. The first presentation of SVM was made by V. Vapnik in [5]. SVM has three main types of classifiers: linear, nonlinear, and C-SVC.

2.2.3. Ensemble methods

An ensemble of classifiers (ensemble method, EM), is a classifier composed of multiple classical models (classifiers) that performs object classification using voting of individual algorithms [7]. To create the base models for each model, the training set is resampled, modified, or randomized. Modifying the learning parameters for individual models is also a common procedure [13]. The conducted theoretical and empirical research confirm the superiority of the effectiveness of ensembles of classifiers over the use of single classifiers [8, 13, 20].

In the case of ensemble methods, we distinguish two main groups: homogeneous and heterogeneous. The first group of approaches is based on using the same classifier but on different distributions of the training set. Within homogeneous methods, bagging, boosting, and random forests [15] are the most popular solutions. For heterogeneous ensembles of classifiers, it is possible to use different underlying algorithms as well as different training set distributions [22].

2.2.4. Bagging

Bagging (bootstrap aggregating) is a method developed by L. Breiman to improve classification, and the performance of regression models for accuracy and stability by reducing variance [2]. It is the first and simplest of the ensemble methods [9].

The bagging algorithm iteratively creates a classifier using a set of bootstrap samples created from the training set. Given a training dataset (of size n), in bagging, at each step (for each individual classifier), a new training pseudo-set of size n is drawn by uniform sampling with full replacement from the original data set. The probability of selecting an element is equal for all instances and is $\frac{1}{n}$. The decisions of the individual classifiers are independent and cast a single vote. The final classification is based on simple voting.

2.2.5. AdaBoost

Boosting is a discrete adaptive learning algorithm which idea is based on creating a "strong and complex classifier" using "weak and simple classifiers". The original concept was refined by Y. Freund and R. E. Schapire and published in [11, 12] as the Adaptive Boosting (AdaBoost) algorithm.

The AdaBoost algorithm starts by assigning the elements of the training set weights ($\frac{1}{n}$, where n is the number of elements in the set) representing the probability of selecting an element into the pseudo-sample. Classification and evaluation of the correctness of the assignment of elements to classes are performed. The weights are then updated. Originally misclassified elements receive higher weights, which allows increasing the probability of selecting an element. The increase depends on the size of the misclassification error, i.e., the sum of weights of elements that were misclassified.

2.2.6. Random forests

Random forest is a method proposed by L. Breiman for constructing an ensemble of classifiers consisting of multiple decision trees that perform the class selection by a simple voting. This approach is very useful in the case of multidimensional data sets, where it allows to increase the accuracy of classification [3].

In the first step, pseudo-samples are generated that will be used to create decision trees. The n elements are drawn for the pseudo-samples from the n -element set. Each element has an equal probability of drawing $\frac{1}{n}$. Using the created pseudo-samples, decision trees are generated. The process is terminated when, for each tree, elements belonging to a single class remain in the node. During tree creation, each split rule is executed independently for different subsets of attributes [3].

3. Prediction of football results based on heterogeneous ensembles of classifiers

The model presented in this paper is a comprehensive KDD approach. All data were collected and processed into a decision table described in Sec. 4.1. Next, analyzed the machine learning algorithms and their results when incorporated into our proposed approach – a heterogeneous ensemble of classifiers. In this section describes the next steps in the implementation of the solution model proposed in this work. The idea is also presented, as Alg. 1.

The first phase of the solution is data preprocessing. Depending on the league to be predicted, the appropriate data is compressed. They go through a selection and transformation process. This produces the cleaned data used to train classifiers. This is indicated in Alg. 1, line 1. In our case, the data collected was limited to 6 conditional attributes and a decision (match result). This is described in detail in Sec. 4.1.

The next step is to prepare a heterogeneous ensemble of classifiers which is a collection of classifiers. The ensemble is defined as follows:

$$HEM = \{d_j : X \rightarrow \{1, 2, \dots, g\}\}_{j=1,2,\dots,J}, \quad (1)$$

where J is the number of different classifiers j ($J \geq 2$).

In the proposed heterogeneous ensemble of classifiers, the predictions of classifiers are combined to make the overall prediction for the ensemble. Classification is done by a voting procedure or only the same decisions are selected (in such an approach the coverage in the classification is incomplete).

For example, in a simple vote, each classifier votes on the decision concerning the sample classification, and the decision with the highest number of votes is chosen. The classifier created by a heterogeneous ensemble of classifiers HEM , denoted as $dHEM : X \rightarrow \{1, 2, \dots, g\}$, uses the following voting rule:

$$dHEM(x) := \arg \max_c N_c(x), \quad (2)$$

where c is a decision class, such that $c \in \{1, 2, \dots, g\}$; and $N_c(x)$ is the number of votes for the sample $x \in X$ classification in to class c , such that $N_c(x) := \#\{j : d_j(x) = c\}$.

The process of preparing the heterogeneous ensemble of classifiers is presented in Alg. 1, lines 2–8. It can be seen there that it involves learning each of the classical classifiers (both single classifiers and ensembles of classifiers) on the collected data. Then, the learned classifier is added to the heterogeneous ensemble of classifiers.

The prepared heterogeneous ensemble of classifiers is used in the next step to predict the outcome of matches that were not used in the learning. The final decision is made (depending on the chosen methodology – more on this in Sec. 4.1) based on simple voting (Eq. (2)) or unanimous predictions of classifiers from the heterogeneous ensemble.

Algorithm 1: Proposed approach

Input: *previous_matches* – historical results of football matches
Input: *matches_to_predict* – matches whose results are to be predicted
Output: *prediction* – prediction of results for new matches

```

1  data_set := data_preprocessing(previous_matches);
2  heterogeneous_ensemble := NULL;
3  // For all classifiers Eq. (1)
4  for i = 1 to J do
5      //classifier[i] – i-th classifier from the ensemble
6      classifier[i] = train_classifier(classifier[i], data_set);
7      add_to_ensemble(heterogeneous_ensemble, classifier[i]);
8  endfor
9  prediction := prediction_by_an_ensemble(heterogeneous_ensemble, matches_to_predict); // Eq. (2)
10 result prediction;

```

4. Experiments

We present some observations resulting from experiments that were performed to test the proposed approach and that allow verifying the research hypothesis. That is, to verify whether the preparation of a relatively simple decision table allows an effective prediction of the results of football matches and whether the use of a heterogeneous ensemble of classifiers allows improving the prediction performed by classical classifiers (including ensembles of classifiers).

To achieve this goal, data were first prepared (Sec. 4.1) and then experiments were performed (Sec. 4.2). In this case, three elements were studied:

- Prediction results obtained by classical algorithms on a prepared decision table, which is simple in analysis.
- Prediction results obtained by the proposed method - heterogeneous ensemble of classifiers (call proposed method (all)).
- Selected prediction enabled by the proposed heterogeneous ensemble of classifiers (call proposed method (selected)).

The results achieved in terms of the quality of classification measures were subjected to appropriate statistical tests. The results of these analyses can be found in Sec. 4.3.

4.1. Experimental design and data set

The proposed approach is related to learning on readily available data. This allows the solution tested on Bundesliga results to be easily scalable to other football leagues for which no complex statistics are kept. In this case, the data set was limited to six attributes:

(1) the number of the current round; (2) the current position of team 1 – in the league table; (3) the current position of team 2; (4) the number of points of team 1; (5) the number of points of team 2; (6) the difference between the points of team 1 and team 2. Where team 1 denotes the home team of the match and team 2 denotes the visiting club. Three decision classes are also designated to represent the outcome of the match: 1 – home team win; 0 – draw; 2 – visiting team win.

Based on preliminary experiments and conclusions from the literature analysis, it was determined that all matches played from the 6th round onwards are used for learning, as the first 5 rounds result in a relatively chaotic distribution in the league table. On the other hand, the entire second round of matches, i.e. 17 rounds from the 2019/2020 Bundesliga season, was used for prediction.

The experiments were performed using the train and test method, dividing the data into a training set (containing all the results of matches from the 6th round in the seasons from 2010/2011 to 2018/2019) and a test set (containing all the matches from the 18th round from the 2019/2020 season). This solution allows both to evaluate the quality of

the classifier and to represent a real situation, in which historical data was used for learning, while the classification is performed on new football matches. The results were evaluated using classical measures of quality of classification.

The resulting training set contains 1827 cases (9 seasons of 29 rounds with 9 games each), while the test set contains 153 cases (17 rounds of 9 games each). The data sets prepared in this way were made available to the public on the website of the University of Economics in Katowice [16]. The division into decision class is presented in Tab. 1 and shows a very similar distribution of decision classes in each of the data sets.

Table 1. Decision classes in training and test sets

Decision class	Counts in the training set	Counts in the test set
1	839 (46%)	71 (46%)
0	440 (24%)	38 (25%)
2	548 (30%)	44 (29%)

The quality of the classification was presented on the classical measures. However, as there are three decision classes in this problem, the values of measures were determined for each of the decision classes (except for the accuracy rate, which is always determined in the same way). All measures that can be determined on the basis of the confusion matrix (example for decision class '1' – Tab. 2): accuracy (Eq. (3)), precision (Eq. (4)), recall (Eq. (5)) and f1-score (Eq. (6)), where c denotes the decision class for which the value of the measure is calculated.

Table 2. Confusion matrix – example for decision class '1'

	Predicted '1'	Predicted '0'	Predicted '2'
Examples '1'	True '1' (TP_1)	False '1' to '0' (FN_1)	False '1' to '2' (FN_1)
Examples '0'	False '0' to '1' (FP_1)	True '0' (TN)	False '0' to '1' (—)
Examples '2'	False '2' to '1' (FP_1)	False '2' to '0' (—)	True '2' (TN)

$$accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (3)$$

$$precision = \frac{TP_c}{TP_c+FP_c} \quad (4)$$

$$recall = \frac{TP_c}{TP_c+FN_c} \quad (5)$$

$$f1 = \frac{TP_c+TP_c}{TP_c+TP_c+FP_c+FN_c} \quad (6)$$

In order to compare the proposed methods, experiments were performed for classical classifiers. On the basis of literature analysis algorithms such as random forest, AdaBoost, and bagging (as an ensemble of classifiers); decision tree (CART), and SVM (as single classifiers) were selected. Exactly the same classifiers were used as part of the heterogeneous ensemble of classifiers. Therefore, the proposed approaches consist of exactly the same classifiers. Therefore, each heterogeneous ensemble of classifiers contains 5 classifiers (random forest, AdaBoost, bagging, CART, and SVM). The exact results of the experiments are presented in Sec. 4.2.

Additionally, the proposed approaches, despite the same way of learning a heterogeneous classifier, differ in their prediction method:

proposed method (all) – works like a classical classifier, determining the decision class for a new object based on a simple vote (Eq. (2)).

proposed method (selected) – in this case, the decision class is assigned only when each of the individual classifiers in the heterogeneous ensemble points to exactly the same class. This method does not guarantee complete coverage of the test set.

The algorithms were implemented in Python 3.8.5 using the scikit-learn library, including the ensemble methods module. While the heterogeneous ensemble of classifiers was implemented from scratch according to Alg. 1. The experiments were carried out on an Intel Core i5 1.60 GHz x 8 Computer with 16 GB RAM.

4.2. Results of the computational experiments

The prediction results of classical algorithms and two proposed solutions are presented in Tab. 3. The values of all previously described measures of the quality of classification for decision classes 1 and 2 are given. Decision class 0 (representing a draw) is omitted because all tested classifiers have serious issues with a draw prediction. Thus, these results were completely absent from the prediction results, despite the fact that its multiplicity is very close to class 2 and is not much lower than in class 1 (see Tab. 1).

By far the best results are obtained for the proposed method (selected), for which most measures of the quality of classification assessment are much higher than all other methods. The accuracy rate is more than 6% better than the next approach (in this case proposed method (all)). The precision for class 1 is also slightly better, while the prediction of outcome 2 is significantly improved (relative to all other methods). In this case, the precision for class 2 increases by 3% – 23% compared to the other methods. Note that for recall for class 2, only bagging is better, which in general for other measures performs worse than other methods. Excluding bagging, the proposed method (selected) is again better than the other methods.

It should be emphasized, that the idea of the proposed method (selected) is to classify only the cases for which all classifiers indicate the same prediction. In this case, the classification coverage decreases. In the conducted experiments it is 0.5229 (the decision was assigned to 80 football matches out of all 153 included in the test set). The other methods have full coverage (precisely 100). This solution is justified by the purpose for which the prediction should be performed. If the classifier would be used to predict all matches, then the proposed method (all) should be analyzed, but if the classifier would be used, for example, to compare against a bookmaker, then a good solution is to consider only selected cases – then the proposed method (selected) is by far the best solution. Again, more attention should be paid to the precision with which it is possible to determine how many relevant objects are in the set of objects classified as relevant.

Table 3. Quality of classification of results 1 and 2 for: single classifiers, ensemble methods, and proposed heterogeneous approaches (the best results in boldface)

	accuracy rate	precision for 1	precision for 2	recall for 1	recall for 2	f1-score for 1	f1-score for 2
CART	0.4902	0.4855	0.5333	0.9437	0.1818	0.6411	0.2712
SVM	0.4967	0.4855	0.6000	0.9437	0.2045	0.6411	0.3051
Bagging	0.4362	0.5366	0.4033	0.5784	0.4030	0.5567	0.4032
Random forest	0.4906	0.4923	0.4819	0.8883	0.2727	0.6335	0.3483
AdaBoost	0.4706	0.4957	0.4118	0.8169	0.3182	0.6170	0.3590
Proposed method (all)	0.4976	0.4901	0.5482	0.9202	0.2455	0.6396	0.3391
Proposed method (selected)	0.5600	0.5478	0.6331	0.9381	0.3247	0.6917	0.4293

In the case of selecting only methods allowing for full coverage of the values of measures, the assessment of the quality of classification is much more spread over each approach. However, the proposed method (all) can be

considered as the most average solution, whose classification accuracy is the highest (excluding the proposed method (selected)), and in the case of statistical tests (Sec. 4.3) has the third mean rank giving way only to the proposed method (selected) and slightly to the SVM algorithm. It is worth noting that many classical classifiers support class 1 (i.e., home team win) because it is more numerous – this results in similar classification accuracy, but much worse precision and recall for class 2 (visiting team win).

4.3. Statistical tests

Statistical tests were determined by means of a non-parametric statistical hypothesis test – the Friedman test for $\alpha = 0.05$. The purpose of conducting tests was to determine if there was a statistical difference between the tested approaches (in terms of all the measures used to evaluate the quality of classification). Additionally, the analyses allow us to determine which method statistically produces the best values of the classification quality.

The application of tests allowed to determining mean ranks for sample determined for each approach, on the basis of classification results evaluated with accuracy, precision (for 1 and 2), recall (for 1 and 2), and f1-score (for 1 and 2) measures. On this basis, it was determined that the best rank, equal to 1.4286 was obtained for the proposed method (version with selection). It is critically better than all other approaches, as the second rank of SVM algorithm is 3.9286 and is 2.5 worse than the proposed method (selected). With a 5% critical difference of just over 2.1.

The second of the proposed methods (version with all cases) was slightly worse than the SVM algorithm (it is 4.000, which means that the difference does not exceed 0.1). The next algorithms have already slightly worse ranks (from 0.43 to 0.79), which, however, does not mean a critical difference.

Table 4. Friedman Test results and mean ranks (the best rank in boldface)

	Values
Chi-Square	12.755784
degrees of freedom	7
p value is less than	0.0471
5% critical difference	2.102509
Mean ranks	
CART	4.7857
SVM	3.9286
Bagging	4.7143
Random forest	4.4286
AdaBoost	4.7143
Proposed method (all)	4.0000
Proposed method (selected)	1.4286

5. Conclusions

The aim of this paper was to develop a new approach for the construction of a heterogeneous ensemble of classifiers used to predict the results of football matches. This was to verify the hypothesis that the application of the developed model based on heterogeneous ensembles in the prediction of Bundesliga match results allows for improved prediction compared to classical ensembles of classifiers and single classifiers. Additionally, the approach based on the prediction of only selected solutions (in which the decision of all classifiers is the same) allows for an even better prediction improvement.

All the goals were achieved so that it was possible to compare the proposed models with classical algorithms. Additionally, the proposed solution based on a simple decision table is possible for basic prediction even by classical algorithms. This approach is scalable and applicable to less popular football leagues than the Bundesliga. The results obtained are significantly higher than a random prediction, as well as random prediction with weights (on the frequency of decision classes). On the other hand, the proposed methods of heterogeneous ensembles of classifiers

further improve the obtained results (as measured by popular measures of the quality of classification evaluation). Statistical analyses show that the proposed method (selected) is critically better than the other approaches. All other measures have no critical differences.

However, there was a serious drawback with the prediction of the draw (class 0). Despite the relatively large number of objects in this class, its prediction is very poor. This is an issue that should be improved in future work. Future work should focus on improving the model (data set construction), using other classifiers in a heterogeneous ensemble of classifiers, and using artificial intelligence methods based on computational intelligence to improve the classification (primarily the mentioned class 0). In the future, the performance of the proposed methods should also be compared with the prediction performed by the bookmakers.

References

- [1] Baboota, R., Kaur, H., 2019. Predictive analysis and modelling football results using machine learning approach for english premier league. *International Journal of Forecasting* 35, 741–755. doi:[10.1016/j.ijforecast.2018.01.003](https://doi.org/10.1016/j.ijforecast.2018.01.003).
- [2] Breiman, L., 1996. Bagging predictors. *Machine learning* 24, 123–140. doi:[10.1007/BF00058655](https://doi.org/10.1007/BF00058655).
- [3] Breiman, L., 2001. Random forests. *Machine learning* 45, 5–32. doi:[10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324).
- [4] Breiman, L., Friedman, J., Stone, C.J., Olshen, R.A., 1984. *Classification and regression trees*. CRC press.
- [5] Cortes, C., Vapnik, V., 1995. Support-vector networks. *Machine learning* 20, 273–297. doi:[10.1007/BF00994018](https://doi.org/10.1007/BF00994018).
- [6] Delen, D., Cogdell, D., Kasap, N., 2012. A comparative analysis of data mining methods in predicting ncaa bowl outcomes. *International Journal of Forecasting* 28, 543–552. doi:[10.1016/j.ijforecast.2011.05.002](https://doi.org/10.1016/j.ijforecast.2011.05.002).
- [7] Dietterich, T.G., 2000. Ensemble methods in machine learning, in: *International workshop on multiple classifier systems*, Springer. pp. 1–15. doi:[10.1007/3-540-45014-9_1](https://doi.org/10.1007/3-540-45014-9_1).
- [8] Džeroski, S., Ženko, B., 2004. Is combining classifiers with stacking better than selecting the best one? *Machine learning* 54, 255–273. doi:[10.1023/B:MACH.0000015881.36452.6e](https://doi.org/10.1023/B:MACH.0000015881.36452.6e).
- [9] Efron, B., 1992. Bootstrap methods: another look at the jackknife, in: *Breakthroughs in statistics*. Springer, pp. 569–593. doi:[10.1007/978-1-4612-4380-9_41](https://doi.org/10.1007/978-1-4612-4380-9_41).
- [10] Eryarsoy, E., Delen, D., 2019. Predicting the outcome of a football game: A comparative analysis of single and ensemble analytics methods, in: *Proceedings of the 52nd Hawaii International Conference on System Sciences*. doi:[10.24251/HICSS.2019.136](https://doi.org/10.24251/HICSS.2019.136).
- [11] Freund, Y., Schapire, R.E., 1997. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of computer and system sciences* 55, 119–139. doi:[10.1006/jcss.1997.1504](https://doi.org/10.1006/jcss.1997.1504).
- [12] Freund, Y., Schapire, R.E., et al., 1996. Experiments with a new boosting algorithm, in: *icml*, Citeseer. pp. 148–156.
- [13] Hansen, L.K., Salamon, P., 1990. Neural network ensembles. *IEEE transactions on pattern analysis and machine intelligence* 12, 993–1001. doi:[10.1109/34.58871](https://doi.org/10.1109/34.58871).
- [14] Joseph, A., Fenton, N.E., Neil, M., 2006. Predicting football results using bayesian nets and other machine learning techniques. *Knowledge-Based Systems* 19, 544–553. doi:[10.1016/j.knosys.2006.04.011](https://doi.org/10.1016/j.knosys.2006.04.011).
- [15] Kozak, J., 2019. Decision tree and ensemble learning based on ant colony optimization. Springer. doi:[10.1007/978-3-319-93752-6](https://doi.org/10.1007/978-3-319-93752-6).
- [16] Kozak, J., Glowania, S., 2021. Bundesliga football results. Data retrieved from, <https://www.ue.katowice.pl/index.php?id=20435>.
- [17] Leung, C.K., Joseph, K.W., 2014. Sports data mining: predicting results for the college football games. *Procedia Computer Science* 35, 710–719. doi:[10.1016/j.procs.2014.08.153](https://doi.org/10.1016/j.procs.2014.08.153).
- [18] Maszczyk, A., Golaś, A., Pietraszewski, P., Rocznik, R., Zajac, A., Stanula, A., 2014. Application of neural and regression models in sports results prediction. *Procedia-Social and Behavioral Sciences* 117, 482–487. doi:[10.1016/j.sbspro.2014.02.249](https://doi.org/10.1016/j.sbspro.2014.02.249).
- [19] McCabe, A., Trevathan, J., 2008. Artificial intelligence in sports prediction, in: *Fifth International Conference on Information Technology: New Generations (itng 2008)*, IEEE. pp. 1194–1197. doi:[10.1109/ITNG.2008.203](https://doi.org/10.1109/ITNG.2008.203).
- [20] Opitz, D., Maclin, R., 1999. Popular ensemble methods: An empirical study. *Journal of artificial intelligence research* 11, 169–198. doi:[10.1613/jair.614](https://doi.org/10.1613/jair.614).
- [21] Pai, P.F., ChangLiao, L.H., Lin, K.P., 2017. Analyzing basketball games by a support vector machines with decision tree model. *Neural Computing and Applications* 28, 4159–4167. doi:[10.1007/s00521-016-2321-9](https://doi.org/10.1007/s00521-016-2321-9).
- [22] Reid, S., 2007. A review of heterogeneous ensemble methods. Department of Computer Science, University of Colorado at Boulder.
- [23] Rotshtein, A.P., Posner, M., Rakityanskaya, A., 2005. Football predictions based on a fuzzy model with genetic and neural tuning. *Cybernetics and Systems Analysis* 41, 619–630.
- [24] Rue, H., Salvesen, O., 2000. Prediction and retrospective analysis of soccer matches in a league. *Journal of the Royal Statistical Society: Series D (The Statistician)* 49, 399–418. doi:[10.1111/1467-9884.00243](https://doi.org/10.1111/1467-9884.00243).
- [25] Schauburger, G., Groll, A., 2018. Predicting matches in international football tournaments with random forests. *Statistical Modelling* 18, 460–482. doi:[10.1177/1471082X18799934](https://doi.org/10.1177/1471082X18799934).
- [26] Schauburger, G., Groll, A., Tutz, G., 2016. Modeling football results in the german bundesliga using match-specific covariates doi:[10.5282/ubm/epub.29390](https://doi.org/10.5282/ubm/epub.29390).
- [27] Schumaker, R.P., Jarmoszko, A.T., Labedz Jr, C.S., 2016. Predicting wins and spread in the premier league using a sentiment analysis of twitter. *Decision Support Systems* 88, 76–84. doi:[10.1016/j.dss.2016.05.010](https://doi.org/10.1016/j.dss.2016.05.010).